

Ekman Layers via Complex Variables

Fluid Dynamics

Homework 10, Problem 2

Selected solution, edited for standalone reading

This excerpt uses a complex velocity to solve the nondimensional, wind-driven Ekman boundary layer in the upper ocean. The vertical coordinate satisfies $z \leq 0$, with the surface at $z = 0$. The velocity perturbation $(\tilde{u}, \tilde{v})$ relative to the geostrophic interior obeys

$$-\tilde{v} = \tilde{u}_{zz}, \quad \tilde{u} = \tilde{v}_{zz},$$

with decay at depth and a unit zonal surface shear:

$$(\tilde{u}, \tilde{v}) \longrightarrow (0, 0) \quad (z \rightarrow -\infty), \quad (\tilde{u}_z(0), \tilde{v}_z(0)) = (1, 0).$$

Complex formulation

Let

$$U = \tilde{u} + i\tilde{v}.$$

From

$$-\tilde{v} = \tilde{u}_{zz}, \quad \tilde{u} = \tilde{v}_{zz},$$

we obtain

$$U'' = \tilde{u}'' + i\tilde{v}'' = -\tilde{v} + i\tilde{u} = iU.$$

Hence, the complex ODE is

$$\boxed{U'' - iU = 0.}$$

The boundary conditions

$$\lim_{z \rightarrow -\infty} \tilde{u} = \lim_{z \rightarrow -\infty} \tilde{v} = 0, \quad \tilde{u}_z(0) = 1, \quad \tilde{v}_z(0) = 0.$$

become

$$\boxed{U(z) \rightarrow 0 \quad (z \rightarrow -\infty), \quad U_z(0) = 1.}$$

Decay and the surface boundary condition

The characteristic equation $r^2 - i = 0$ gives

$$r = \pm\lambda,$$

where

$$\lambda = \sqrt{i} = e^{i\pi/4} = \frac{1+i}{\sqrt{2}}.$$

Thus

$$U(z) = Ae^{\lambda z} + Be^{-\lambda z}.$$

Decay as $z \rightarrow -\infty$ requires $B = 0$, and $U_z(0) = A\lambda = 1$, so

$$A = \frac{1}{\lambda}, \quad \boxed{U(z) = \frac{1}{\lambda}e^{\lambda z}.$$

Vertically integrated transport

$$\tilde{U} = \int_{-\infty}^0 U dz = \frac{1}{\lambda} \int_{-\infty}^0 e^{\lambda z} dz = \frac{1}{\lambda^2}.$$

Since $\lambda^2 = i$,

$$\boxed{\tilde{U} = \frac{1}{i} = -i.}$$

Velocity components and physical interpretation

Write

$$\lambda = \frac{1+i}{\sqrt{2}}.$$

Then,

$$U(z) = \frac{1}{\sqrt{2}}(1-i)e^{\frac{(1+i)}{\sqrt{2}}z},$$

which gives

$$\begin{aligned}\tilde{u}(z) &= \frac{e^{z/\sqrt{2}}}{\sqrt{2}} \left[\cos\left(\frac{z}{\sqrt{2}}\right) + \sin\left(\frac{z}{\sqrt{2}}\right) \right], \\ \tilde{v}(z) &= \frac{1}{\sqrt{2}} e^{\frac{z}{\sqrt{2}}} \left(\sin\left(\frac{z}{\sqrt{2}}\right) - \cos\left(\frac{z}{\sqrt{2}}\right) \right).\end{aligned}$$

From $\tilde{U} = -i$,

$$\int_{-\infty}^0 \tilde{u} dz = \Re(\tilde{U}) = 0, \quad \int_{-\infty}^0 \tilde{v} dz = \Im(\tilde{U}) = -1.$$

Thus the vertically integrated perturbation flow is orthogonal to the purely zonal wind stress.